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RESEARCH INTERESTS

Single-cell clustering algorithms grounded in statistical geometry, mathematical optimization, and AI/ML: structure-preserving embeddings, graph- and mixture-based cluster inference, and scalable models for high-dimensional single-cell data.

EDUCATION

Wright State University, Dayton, OH Jan 2026 – Present

Ph.D. in Computer Science Focus: HDLSS Biological Data, Dimensionality Reduction, Healthcare AI

Wright State University, Dayton, OH Jan 2023 – Dec 2024

M.S. in Computer Science

National Institute of Technology, Uttarakhand, India Jul 2018 – May 2022

B.Tech. in Electronics and Communications Engineering

RESEARCH EXPERIENCE

Graduate Research Assistant, Wright State University Jan 2026 – Present

- Studying the centroid-encoder family of linear/nonlinear supervised dimensionality-reduction methods (Centroid-Encoder, OCSR, CBOCSR, FisherLDA, LFDA, SELF) and the Sparse Linear Centroid-Encoder (SLCE) biomarker-selection model, validated on high-dimensional, low-sample-size (HDLSS) gene-expression datasets (ALLAML, GLI_85, Pancan), with downstream applications such as single-cell clustering and biomarker discovery for healthcare and biomedical research.
- Co-developed an isometry-constrained variant of the Centroid-Encoder to preserve pairwise distance structure during projection of HDLSS biological data, forming the basis of the manuscript listed under Publications.
- Extending centroid-encoder validation to new miRNA expression datasets via a Supervised Centroid Encoder (SCE) pipeline: a 90-subject muscle exercise-response study (four timepoints pre-exercise, 24h post, 12-week 24h post, 16-week rest) and a serum miRNA dataset, evaluating embedding quality with kNN accuracy, Kruskal cost, stability, and trustworthiness metrics.
- In parallel, conducting research on Generative AI and AI Assurance for safety-critical computer vision systems architecting testing frameworks, generative synthetic-data pipelines, adversarial stress tests, digital-twin simulation, and agentic LLM evaluation loops (generate → test → critique → refine) for threat-detection and autonomous-system applications.

Independent Research & Development Projects 2026 (ongoing)

- Conducting a structured study of imputation methods for high-dimensional, low-sample-size omics data across five mathematical families: probabilistic/empirical-Bayes (SAVER, scImpute), low-rank/spectral (softImpute, ALRA), diffusion/graph-based (MAGIC), iterative conditional (MICE, missForest), and optimal transport (Wasserstein barycenters, Sinkhorn regularization).
- Extending the analysis to multi-omics cross-modal imputation (MOFA+, totalVI) and covariance/precision-matrix shrinkage (Ledoit-Wolf/OAS, Graphical LASSO) under $p \gg n$ regimes.
- Building a failure-mode taxonomy, the precise geometric assumption each method makes and where it breaks as the foundation for designing a new, geometry-respecting imputation algorithm (optimal transport as the unifying mathematical framework).
- Designed and trained Variational Autoencoders (input → 400 → latent $[\mu, \sigma, \text{dim}=50]$ → 400 → output; MSE reconstruction + KL-divergence loss) to generate synthetic samples for two small-n microarray datasets: MOSSD (Affymetrix miRNA 4.0, ~6,631 miRNA probes, 29 human samples) and Susuki (Clariom S Mouse, ~22,206 genes, 10 mouse samples).
- Validated synthetic-sample fidelity by comparing PCA projections of original vs. VAE-generated data in both raw and normalized space, to assess whether generative augmentation preserves biological structure in extreme HDLSS regimes ($n \leq 30$).

Graduate Coursework Project RNA-seq Pipeline for Host–Pathogen Transcriptomics *Spring 2026*

- Built and ran the primary end-to-end RNA-seq pipeline (SRA accession → FASTQ → alignment → gene-level quantification) for PRJNA938511 (Ebola virus RNA-seq in rhesus macaques; 356 SRR runs / 308 biological samples) on the Ohio Supercomputer Center (OSC) HPC cluster using SLURM array jobs.
- Retrieved and converted raw reads with the SRA Toolkit (prefetch, fasterq-dump) and pigz/gzip compression; performed adapter and UMI trimming and quality control with fastp and FastQC.
- Constructed a hybrid *Macaca mulatta* (Mmul10) + EBOV (KJ660346.2) reference genome and ran two-pass splice-aware alignment with STAR (v2.7.11b), followed by gene-level counting with HTSeq-count.
- Assembled and validated a 35,439-gene × 308-sample count matrix in R, reproducing the original GEO submission (GSE226106) via custom validation scripts and resolving a gene-ordering discrepancy through ID-based joins.
- Performed exploratory data analysis (PCA/MDS, DESeq2 variance-stabilizing transformation, sample-distance clustering) to confirm biologically coherent sample structure across tissues and timepoints.
- Co-authored the unified project report and LaTeX write-up with a lab collaborator, whose independent Kallisto/HISAT2 viral-transcriptome pipeline cross-validated the host-transcriptome counts (Pearson $r = 0.9978$).
- **Tools & technologies:** SRA Toolkit, fastp, FastQC, STAR, HTSeq, R (DESeq2), Python, Bash/SLURM, OSC HPC cluster, LaTeX.

PUBLICATIONS

Published:

- Changala, S.R.K.P. et al. “Stable Dimensionality Reduction via Isometric Centroid-Encoder.” ResearchGate.

TECHNICAL SKILLS

- **AI/ML:** Generative AI, Computer Vision, LLMs, Agentic AI, Adversarial ML, Synthetic Data Generation, AI Assurance, PyTorch, TensorFlow
- **Languages:** Python, RScript, Java, JavaScript/TypeScript, C++, SQL
- **Cloud/Infra:** AWS, Kubernetes, Docker, Oracle Cloud (APEX), Git
- **Data:** Neo4j (Graph Databases), SQL

CERTIFICATIONS & TRAINING

- Machine Learning Specialization - Stanford / DeepLearning.AI, Coursera (2024)
- Neo4j Certified Professional - Neo4j (2025)
- AWS Certified Cloud Practitioner - Amazon Web Services (2024)
- AWS Cloud Quest: Serverless Developer - Amazon Web Services
- Oracle APEX Cloud Developer Certified Professional - Oracle (2025)
- GitHub Foundations - GitHub

PRIOR ACADEMIC PROJECTS (M.S., 2023–2024)

- Pneumonia Detection System: CNN/RNN pipelines classifying chest X-ray images, optimizing inference time and accuracy.
- Weather Image Classification: 11-class classification over 6,862 images; exploratory data analysis and feature visualization.
- Cloud-based TSP Solver: Deployed on AWS EC2 and Ohio Supercomputer Center nodes with a hosted web frontend.
- Information Retrieval: Built indexing, web crawling, query processing, and ranking algorithms.

OTHER PROFESSIONAL EXPERIENCE

Full Stack / Java Developer, AcentrikTech LLC, Austin, TX

Mar 2025 – Dec 2025

Full Stack Web Developer, Edureka, Bengaluru, India

Feb 2022 – Aug 2022